

BCT

What is shared digital economy?

Shared digital economy is a concept that leverages online technology and digital platforms to enable users to share resources and services in an efficient and collaborative manner. It includes activities such as peer-to-peer sharing, collaborative consumption, and the sharing of digital goods and services. Examples of shared digital economy activities include ride-sharing, collaborative consumption of goods and services, and the sharing of digital products such as e-books and software.

Write briefly on at least 4 emerging technologies.

1. Artificial Intelligence (AI): AI is the science of teaching machines to think and act like humans. AI is used in a variety of applications, such as natural language processing, computer vision, robotics, and autonomous vehicles.
2. Augmented Reality (AR): AR combines the real and virtual worlds to create an enhanced environment where users can interact with digital objects. Examples include virtual reality goggles, augmented reality apps, and augmented reality games.
3. Internet of Things (IoT): IoT is a network of connected devices that can collect, exchange, and act on data. IoT is used in applications such as smart homes, smart cities, and industrial automation.
4. Blockchain: Blockchain is a distributed ledger technology that enables secure and transparent peer-to-peer transactions. It is used in applications such as cryptocurrency, smart contracts, and distributed data storage.

What is cloud? What is the difference between hybrid cloud and multi-cloud?

Cloud computing is a type of computing that utilizes a network of remote servers hosted on the Internet to store, manage, and process data, rather than a local server or a personal computer. It provides access to networks, servers, storage, databases, and a broad set of application services over the Internet.

A hybrid cloud is a cloud computing environment that utilizes a combination of both public and private clouds, allowing for data and applications to be shared between the two.

A multi-cloud is a cloud computing environment that utilizes multiple cloud computing services from different cloud providers. This allows for data and applications to be distributed across multiple clouds for added security, scalability, and reliability.

explain 3-tier architecture. What are the cloud services that helps in building three tier architecture.

3-tier architecture is a type of software architecture that is composed of three “tiers” or layers of logical computing. The three tiers include the presentation tier (client/user interface), the application tier (business logic and data access), and the data tier (data storage and management). Each tier is separate and distinct and is designed to handle specific application functions.

The cloud services that help in building three-tier architecture include Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS). IaaS provides the physical or virtual infrastructure, such as virtual machines, storage, and networking, which the other tiers can be built on. PaaS provides a framework and runtime environment for building and running applications, including development and deployment tools. SaaS provides a complete application solution with everything from user interfaces to data storage and management.

Presentation tier

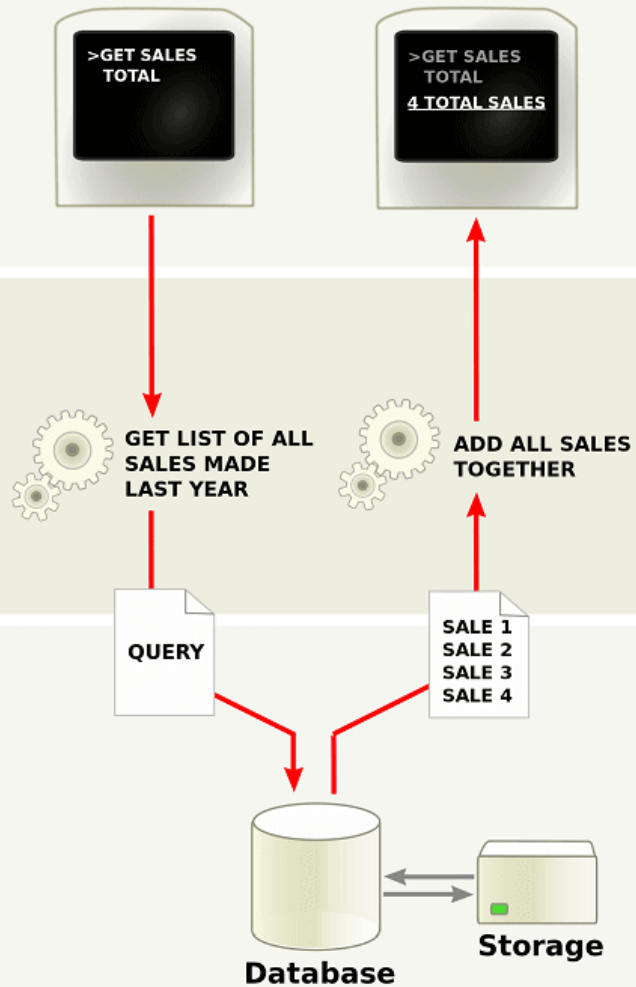
The top-most level of the application is the user interface. The main function of the interface is to translate tasks and results to something the user can understand.

Logic tier

This layer coordinates the application, processes commands, makes logical decisions and evaluations, and performs calculations. It also moves and processes data between the two surrounding layers.

Data tier

Here information is stored and retrieved from a database or file system. The information is then passed back to the logic tier for processing, and then eventually back to the user.



What is cryptography? Why study of cryptography is very key to BCT?

Cryptography is the practice of using complex mathematical algorithms to encrypt and decrypt data. It is used to protect data from being accessed by unauthorized individuals. The study of cryptography is very important for BCT because it is used to protect sensitive information, such as financial transactions, from being accessed by unauthorized individuals. Cryptography also helps to ensure that data is not altered or tampered with. By studying cryptography, BCT professionals can develop secure systems that protect confidential data and ensure data integrity.

What is digital signature? Explain its's advantage.

A digital signature is an electronic, encrypted, signature that can be used to authenticate the identity of the sender of a digital message or document. It is used to ensure that the message or document came from the stated sender and has not been altered in transit. Digital signatures provide a higher level of security than traditional physical signatures and can help to prevent fraud. The main advantage of using digital signatures is that they provide assurance and security

that the message or document came from the stated sender and has not been altered in transit. Digital signatures also provide non-repudiation, meaning that the sender cannot deny having sent the message or document. Digital signatures are also more convenient and cost-effective than traditional physical signatures.

What is a ledger? List top 5 DLT

A ledger is an organized record of financial transactions.

1. Ethereum: Ethereum is a decentralized platform that runs smart contracts. It enables developers to build and deploy decentralized applications (dApps).
2. Hyperledger Fabric: Hyperledger Fabric is an open source blockchain platform for businesses. It is a permissioned network that offers plug-and-play components and a modular architecture.
3. R3 Corda: R3 Corda is a distributed ledger technology (DLT) platform designed for businesses. It enables businesses to securely and efficiently transact with each other.
4. Quorum: Quorum is a distributed ledger technology (DLT) platform developed by JPMorgan Chase. It is designed to enable enterprises to securely and privately transact with each other.
5. EOS: EOS is a blockchain-based platform designed to enable developers to build, deploy, and run decentralized applications (dApps). It is built on a Delegated Proof of Stake (DPoS) consensus mechanism.

1. Observe a real-life ecosystem to spot a problem that can be solved using Blockchain technology. Explain your solution and highlight the advantages of your solution.

One real-life problem that could potentially be solved using blockchain technology is supply chain traceability and transparency. Many supply chains, particularly those involving physical goods, can be complex and involve multiple intermediaries, making it difficult to track the origin and movement of products. This can lead to issues such as counterfeiting, fraud, and unethical practices, such as using child labour or exploiting workers.

A solution to this problem could be to use a blockchain-based system to track and verify the movement of goods through the supply chain. Each step in the supply chain could be recorded on the blockchain, creating a transparent and immutable record of the history of the product. This would allow consumers and businesses to

easily trace the origin and movement of products, ensuring that they are legitimate and ethically produced.

There are several advantages to this solution:

1. Increased transparency: Blockchain technology allows for a transparent and verifiable record of the supply chain, enabling consumers and businesses to easily track the origin and movement of products.
2. Improved traceability: The immutability of the blockchain means that the supply chain record cannot be altered, ensuring that it is accurate and reliable.
3. Reduced counterfeiting and fraud: By creating a transparent and verifiable record of the supply chain, it would be more difficult for counterfeits or fraudulent products to enter the market.
4. Improved ethics: By making it easier to trace the origin and movement of products, it would be easier to identify and address any unethical practices in the supply chain.
5. Increased efficiency: A blockchain-based supply chain system could reduce the need for intermediaries and streamline the movement of goods, increasing efficiency and reducing costs.

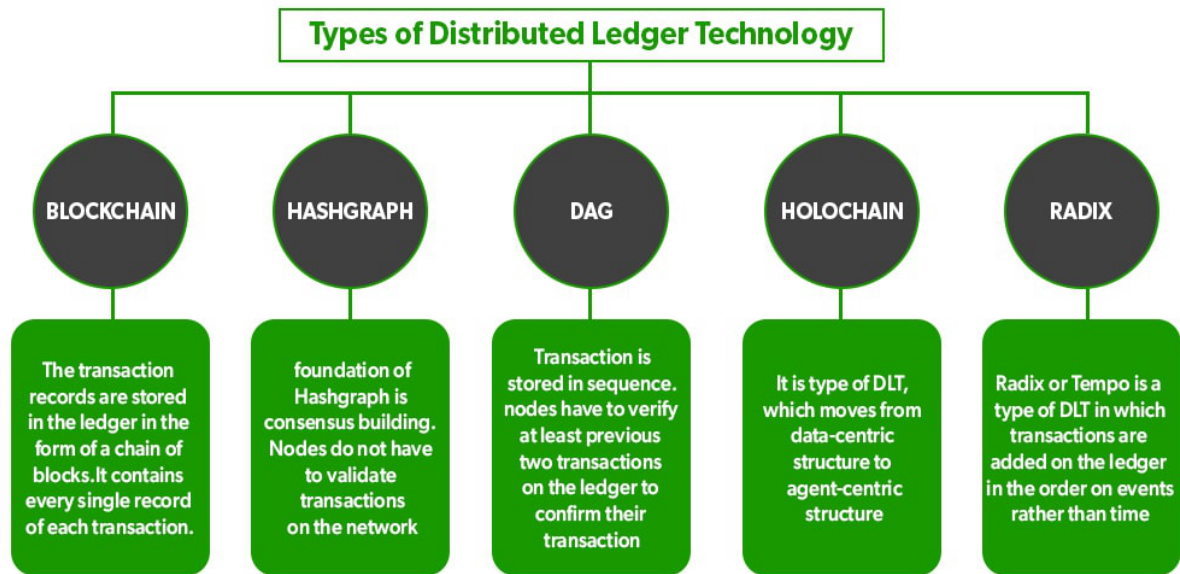
1. Explain with a window diagram the DLT permission categories.

Distributed ledger technology (DLT) is a decentralized system for storing and managing data, such as transactions or records. DLT systems often have different categories of permissions that determine who can access and modify the data on the ledger. Here is a general overview of some common permission categories in DLT systems:

Public: Anyone can access and contribute data to the ledger. This is common in open, permissionless DLT systems, such as Bitcoin or Ethereum.

Private: Only a select group of participants, such as a consortium of banks, have access to the ledger. This is common in closed, permissioned DLT systems.

Hybrid: A combination of public and private access, where certain data is available to the public and other data is restricted to a select group of participants.



2. Explain diagrammatically three types of ledger. Specify one advantage and disadvantage of each of these three ledgers.

1. Centralized Ledger:

Advantage: Easier to manage and control.

Disadvantage: Data can be easily manipulated and is vulnerable to cyber-attacks.

2. Decentralized Ledger:

Advantage: More secure and reliable due to its distributed nature.

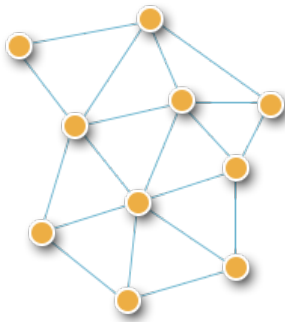
Disadvantage: Difficult to manage and control.

3. Distributed Ledger:

Advantage: Highly secure due to its distributed and decentralized nature.

Disadvantage: Transactions may take longer to process and be more expensive.

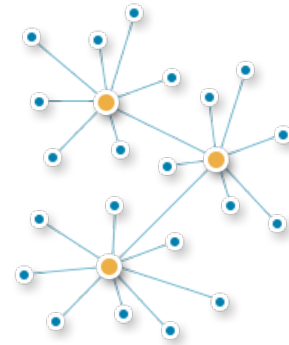
Distributed



Centralized



Decentralized



1. What is Block-chain Explain your understanding of immutability in block chain. How block chain overcomes “double spending” issues.

Blockchain is a distributed ledger technology (DLT) that allows for the secure and transparent recording of transactions or data. It consists of a chain of blocks, each of which contains a record of multiple transactions. The blocks are linked together in a chronological order, forming a chain, and are secured using cryptographic techniques.

Immutability refers to the inability to change or modify something. In the context of blockchain, it means that once a block of transactions has been added to the chain, it cannot be altered or deleted. This is achieved through the use of cryptographic hashes, which are unique digital fingerprints that are created for each block. If any data within a block is changed, the hash of that block will also change, making it easy to detect any tampering.

One of the main benefits of immutability in blockchain is that it helps to ensure the integrity and security of the data on the chain. It makes it difficult for anyone to alter or delete data without being detected, which is important for maintaining trust in the system.

Blockchain technology also helps to overcome the issue of "double spending," which refers to the ability to spend the same digital currency or asset more than once. In a traditional electronic payment system, it is possible for a user to make multiple copies of a digital currency and spend it multiple times. However, in a blockchain system, each transaction is recorded on the chain and validated by multiple network participants. This makes it difficult for a user to spend the same digital currency or asset more than once, as the network would recognize the attempt and prevent it from occurring.